

If a Statistical Model Predicts That Common Events Should Occur Only Once in 10,000 Elections, Maybe it's the Wrong Model*

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Abstract

Political scientists forecast elections, not primarily to satisfy public interest, but to validate statistical models used for estimating quantities of scholarly interest. Although scholars have learned a great deal from these models, they can be embarrassingly overconfident: Events that should occur once in 10,000 elections occur almost every other election cycle, and even those that should occur once in 5 quintillion elections are sometimes observed. We show that these problems miss parts of the predictive distribution of central importance to political science, such as by underestimating how often popular incumbents lose their seats. We then build the first generative statistical model of all US federal legislative elections, covering both the U.S. House (1946–2024) and Senate (1954–2024), and validate it with extensive out-of-sample tests. It turns out that the same set of structural parameters forecast accurately in both houses of congress, potentially implying a single data-generation process for all national legislative elections in the postwar era. This model suggests the possibility of more reliable estimation of quantities of scholarly interest, in addition to accurate forecasts.

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1 Introduction

Political scientists have studied democratic elections for most of the history of our discipline, producing an extensive, high quality, and steadily improving scholarly literature with few equals. Statistical studies of district-level election returns — estimating descriptive and causal effects, counterfactual quantities, forecasts, and full generative models of numerous phenomena — supplemented by a wide variety of other approaches — such as intensive interviews, survey research, participant observation, archival work, and historical analyses — have produced an enviable record of reliable knowledge about the workings of this crucial democratic institution.

Yet, our commonly used statistical models are often spectacularly wrong. This is easiest to see in election prediction, where rigorous out-of-sample evaluations are unforgivingly obvious, and a major concern even when prediction is not the immediate goal. Although standard models do remarkably well much of the time, and have taught us a great deal, they are embarrassingly far off with regularity. These mistakes are not ordinary errors of ordinary magnitudes. Our best models indicate that certain events we see regularly should be rarely observed even if we had data from a trillion elections and, as we show, some from even 5 quintillion elections.

The intrepid political scientists who give media interviews after elections take one for our team trying to explain this to the public (including the authors of this paper). But pretty much the best we can do is to say something like “Oops!... We Did It Again” and to explain that voters get to cast ballots for whomever they want. However, we all know (to paraphrase Britney Spears again) we’re not that innocent. Errors of such magnitude are not merely mistakes. They are bugs in our logic, our models, our forecasts, our conclusions, our textbooks, our advice, and our public pronouncements — similar to what we would think if we built a computer program to forecast the Democratic vote proportion, hit run, and it played a video of a galloping giraffe. This is not a missed forecast; it’s the wrong model. Even the wrong models can produce valuable scientific knowledge for those parts that still fit well — such as for simple quantities like certain means or coefficients, which we show work well in existing approaches — but models that do so badly

when they are vulnerable to being proven wrong, as in prediction problems, do not inspire confidence for tasks more difficult to evaluate, such as causal inferences or generatively accurate descriptive summaries.

In this paper, we build a generative statistical model that is both consistent with the data and capable of estimating many of the diverse quantities political scientists find of interest. These include descriptive quantities — such as the probability of an incumbent losing, the odds of a competitive election, the expected vote of the median house seat, partisan bias, electoral responsiveness, among others — and, with appropriate additional assumptions, causal and other counterfactual inferences. Only a generative model can provide sufficient generality to estimate all these and other quantities, along with accurate uncertainty estimates, which is unlike approaches better for more specific purposes, such as via model-free, distribution-free, machine learning, or semi-parametric approaches. Such a model should also be capable of making election forecasts, but we (and most other political scientists) are not primarily interested in forecasting except as citizens to participate in the discussion leading up to an election. After all, from an academic perspective, the best method of forecasting is well known: just wait a bit. However, ensuring that we have a useful generative model requires that it be made vulnerable to being proven wrong in as many ways as possible, for which forecasting — along rigorous out-of-sample evaluations — is essential.

We build on the long history of Bayesian hierarchical and multilevel modeling in political science, including studies of electoral districts (Gelman and King, 1994; Katz and King, 1999), countries (Western, 1998), state-level opinion from national polls (Park, Gelman, and Bafumi, 2004; Lax and Phillips, 2009), legislators’ ideal points (Martin and Quinn, 2002; Clinton, Jackman, and Rivers, 2004), coefficients that vary over time and space (Shor et al., 2007; Beck and Katz, 2007), individual support for the EU within parties within states (Steenbergen and Jones, 2002), and forecasts that pool state polls over a campaign (Linzer, 2013). See also the general methodological treatments (Jackman, 2009; Gill, 2008; Goplerud, 2022).

We describe the standard modeling approach and our proposed alternative in Section

2, perform many out-of-sample evaluations in Section 3, and extend our results to the US Senate in Section 4. Section 5 discusses the vast array of quantities of interest that can be computed from generatively accurate models, such as the one we now introduce. Details appear in a separate Supplementary Appendix.

2 Statistical Models of District-Level Elections

We first summarize the standard model used in the literature (Section 2.1) followed by our proposed alternative (Section 2.2) designed to incorporate more elections, more substantive knowledge of elections, and more of the known statistical issues with these data, all within a single Bayesian model.

2.1 Standard

The outcome variable for modeling US congressional elections is the Democratic proportion of the two-party vote, v_{it} for district i and election year t . The standard model is a linear-normal regression of v_{it} on a vector of K covariates X_{it} , with estimation conducted for each election year t run independently. For most applications in the last quarter century, an independent normal district-level random effect, γ_i , (constant over hypothetical or real elections but varying over districts) is added to the regression to model the political uniqueness of individual districts (Gelman and King, 1994, implemented in JudgeIt software).¹

The specific content of the covariates varies by application but, to fix ideas and for the analyses below, define X_{it} to include a lagged vote share ($v_{i,b(t)}$, Table S1), incumbent party (the party that won the previous election, with 1 for Democrat and 0 for Republican), incumbency status (1 if the Democratic candidate is an incumbent, 0 for open seat, and -1 for a Republican incumbent), uncontestedness (1 if a Democrat runs uncontested, 0 if contested, and -1 if Republican runs uncontested), and an indicator for the old confederate states.

¹Instead of directly estimating γ_i and modeling multiple elections together, which would have been computationally difficult in the 1990s, JudgeIt analyzes one election at a time after a preprocessing step to estimate how much variation should be attributed to this random effect.

We summarize the standard model as

$$\begin{aligned} v_{it} &\sim \mathcal{N}(\mu_{it}, \sigma^2) \\ \mu_{it} &= X_{it}\beta_t + \gamma_i \end{aligned} \tag{1}$$

where β_t is a vector of K linear regression effect parameters, $\gamma_i \sim \mathcal{N}(0, \sigma_\gamma^2)$ is an independent normal random effect with variance $\sigma_\gamma^2 > 0$, and σ^2 is the variance of the usual homoskedastic regression independent normal error term.

2.2 Proposed

We now build on the standard model to develop our proposed approach, keeping the same flexibility in the choice of covariates within a fully Bayesian framework. We designed each feature of the model using the field’s knowledge of elections unearthed from a venerable literature, in six steps. Section 2.2.1 provides a qualitative description of model components that reflects knowledge from this literature but which has been excluded from the standard approach. Section 2.2.2 combines these components into a single Bayesian model, focusing, for expository purposes, on the simple special case where all elections are contested. Section 2.2.3 extends the model further by parameterizing the uncertainty structure and degree of electoral concentration. Section 2.2.4 extends the model to allow for uncontested elections and Section 2.2.5 expands the model to include both the House and Senate from one data generation process. Lastly, Section 2.2.6 gives our default priors.

The resulting model, which we call “LogisTiCC” for the additive logistic t model with contemporaneous correlations, fits all federal legislative elections with 313 structural parameters and 21,528 random effects (15,973 candidate effects and 5,555 latent lagged votes), most of which are shrunk towards 0 or integrated over. (To verify the essential role of each component included, Supplementary Appendix S2 reports on exhaustive “ablation tests” we conducted that remove one part of the model at a time and observe the change in out-of-sample forecasting performance. In every case, removing a feature degrades performance, confirming the need for the entire model as presented here. Supplementary Appendix S3 runs the test in reverse, adding the components to the standard model one at

a time, and each addition improves held-out forecasts of House elections.)²

2.2.1 Novel Model Components

Error terms in statistical models are designed to represent “known unknowns,” features that reflect political scientists’ knowledge of elections too difficult to code in the covariates. For example, the error term in Equation 3 allows for *candidate uniqueness* by adding a random term for each major-party candidate, $\gamma_{D(i,t)}$ and $\gamma_{R(i,t)}$, where $D(i,t)$ and $R(i,t)$ are the Democratic and Republican candidates in district i at election t , to model the persistence of this uniqueness for any one candidate over time, beyond changes due to X . For example, Minnesota’s 7th Congressional District has long been more Republican than the nation as a whole, favoring Donald Trump in 2016 and 2020 by about 30 percentage points. Yet, Democrat Collin Peterson won this seat from 1991 to 2021 because of his personal brand and unusual political preferences, opposing abortion and supporting the border wall, but (perhaps accounting for how he wins the Democratic nomination) highly progressive economic views.

We now add to this model four other “known unknowns,” modeling features that reflect valuable substantive political information well understood by students of elections or observable in the data but rarely modeled directly. First is *parameter drift with coefficient stability*: β_t varies a lot in the long run, but changes very little from year-to-year. For example, the incumbency advantage might range between two and ten percentage points, with only rare sharp changes over time. Similarly, the coefficient on the lagged vote is usually in the range of $[0.6, 0.8]$. We add this feature to the model by assuming that the time-varying elements β_{tk} of vector β_t (corresponding to covariate k , $k = 1, \dots, K$ and time t) come from a random walk, $\beta_{tk} \sim \mathcal{N}(\beta_{t-2,k}, \sigma_{\beta_k}^2)$, where $\sigma_{\beta_k} < \infty$ is a standard deviation for the year-to-year variation.³ The idea here is to borrow strength for the esti-

²One of the reasons our generative modeling approach has not been tried before is that it would have been computationally infeasible even a few years ago. We designed new algorithms especially for recent hardware and are able to complete one run on eight decades of congressional elections in about 11 GPU hours; adding calibration and full out-of-sample evaluation requires about 73 GPU hours of run time, producing about 630gb of output. Along with this paper, we are making available easy-to-use open source software that implements all our algorithms and methods. See Supplementary Appendix S1 for the full likelihood and Supplementary Appendix S4 for computational details.

³In contrast, estimating each equation separately and independently, as in the standard approach, is equivalent to setting $\sigma_{\beta_k} \rightarrow \infty$. In addition, the notation β_{t-2} is shorthand for a random walk, meaning that

mate of each parameter in each year from the estimation of the same parameter in other years, but without the rigidity and potential bias that would come from a more “informative” prior. This will be especially valuable in smaller legislatures, such as for elections in the US Senate.

Second, we allow for *positive cross-district covariances* by adding a random national trend term, η_t , that allows all districts in one election to be affected in roughly the same way by the same national event, over and above the information in X . For example, the 1994 Republican national congressional campaign strategy (known as the “Contract With America”) seemed to be a successful heresthetical maneuver (Riker’s term for winning by changing the terms of an electoral contest, as opposed to persuading the voters; Riker 1990; Shepsle 2003) that moved all the districts in the Republican direction by approximately the same amount. Although we usually do not know ex ante what any one national trend will be, we can estimate the variation caused by the national trends, which we know occur regularly, and represent this uncertainty in the model with a common random effect for all districts. The result is the well known “approximate uniform partisan swing” pattern that can be found across time periods, electoral systems, and countries (Katz, King, and Rosenblatt, 2020).

Third, we model *district-level political surprises*, including intentional heresthetical maneuvers and unintentional exogenous political events that affect one district’s vote at a point in time differently than others and are not included in X . We explicitly model the parameters that determine these surprises. Consider for example the election in Texas’ 22nd district in 2006. Tom DeLay was the popular Republican House majority leader from the district, regularly winning election by 20 or more percentage points. During the campaign, he was indicted and abruptly resigned. Worse for his party, the deadline to field a candidate on the ballot line had passed and so his party could only field a write-in candidate late in the campaign. The result was that this overwhelmingly Republican district elected a Democrat over the Republican write-in candidate by over 8 percentage points. Equation 1 already includes the usual normal error term that can be used to model

this distribution shrinks different covariate effects towards the previous election’s without favoring one’s a priori guess; this is equivalent to a fully Bayesian model with the mean in the null space; see Girosi and King 2008.

surprises, but a normal distribution indicates that deviations from a prediction this large would happen so infrequently that it would almost never be observed. Of course, as every election observer is aware, these surprises happen regularly, even if we do not know which ones will occur. As we explain below, we will therefore swap out the normal distribution for one that can more appropriately represent these political surprises, also keeping predictions within the (0,1) interval.

Finally, the normal distribution used in the standard approach, which often works well when the mean or average vote outcome is of interest, is inadequate for two more specific reasons. The first reason is that it fails dramatically for most other aspects of the distribution, such as for uncertainty estimates or the probability of a close election or of one party winning. The second reason is that the normal tail implies that big surprises should almost never occur, meaning that it also gets the concentration around the mean wrong. To fix these problems, we use the additive logistic Student t (ALT) distribution, which, unlike the normal, constrains the vote proportion to the (0,1) interval (which must be true) and also has appropriately fatter tails to represent surprises. In addition, the ALT distribution has the simultaneous advantage of having more of its density concentrated near the mean, making the mean (and the covariates that account for its variation) more informative at the same time as it is accounting better for surprises.⁴ The ALT distribution thus allows more informative predictions to coexist in the same model with the possibility of huge surprises. Supplementary Appendix S5 describes alternatives we tried and rejected, including Student- t , additive logistic normal, Beta, and mixture distributions.

2.2.2 The Model, with Fully Contested Elections

We now combine all the features described above in one model (redefining symbols from Section 2.1). Our structural covariates, X_{it} , are $y_{i,b(t)}^*$, I, S, L^{pres}, U, O^{inv}, V, A_d, E_d, L^{sen}, L^{gov} and M. We fully define them in Table S2. For expository purposes, consider the simple case where all districts are contested. The usual single error variance is decomposed by the ALT, for additional flexibility, into scale ϕ_{it} and degrees of freedom ν_{it} (the Greek

⁴The ALT is the implied distribution on v (and so restricted to the (0,1) interval) when the t distribution is applied to the (unbounded) logistic transformation of the vote $\ln[v_{it}/(1 - v_{it})]$. For technical details, and extensive evaluations, see Katz and King (1999).

letter Nu) parameters, each indexed by the election (as $\nu_{it} \rightarrow \infty$, the ALT approximates the additive logistic normal). Thus, let

$$v_{it} \sim \text{ALT}(\mu_{it}, \phi_{it}, \nu_{it}), \quad (2)$$

$$\mu_{it} = X_{it}\beta_t + \gamma_{D(i,t)} - \gamma_{R(i,t)} + \eta_t, \quad (3)$$

$$\phi_{it} = \exp(\zeta + \zeta_d + Z_{it}\delta), \quad (4)$$

$$\nu_{it} = 1 + \exp(\alpha + \alpha_d + Z_{it}\lambda), \quad (5)$$

with indices i for district, t for election year, and d for redistricting decade, and where Z_{it} is a vector containing covariates that predict dispersion—an open-seat indicator, the marginal distance from 50/50, $|\text{logit}^{-1}(L^{\text{pres}}) - \frac{1}{2}|$, and an indicator for single-district states and the Senate—with slopes δ and λ . We also include intercepts ζ and α , with redistricting-decade fixed-effects ζ_d, α_d to model secular changes in these parameters over time in a computationally tractable fashion. The systematic component for the conditional expected value includes three hierarchical terms, for the time-varying covariate effects, candidate, and the national trend, respectively:

$$\beta_{tk} \sim \mathcal{N}(\beta_{t-2,k}, \sigma_{\beta_k}^2), \quad \gamma_c \sim \mathcal{N}(0, \sigma_\gamma^2), \quad \eta_t \sim \mathcal{N}(0, \sigma_{\eta, d(t)}^2)$$

for candidates c and election years t in redistricting decade $d(t)$.

The candidate effects $\gamma_{D(i,t)}$ and $\gamma_{R(i,t)}$ are person-specific intercepts on the Democratic and Republican candidates in the race, drawn from a common distribution; both identities are fixed at candidate selection, before the outcome, and a race in which one major party fields no candidate drops that party's term. Each is the candidate's valence net of party, incumbency, and the district's partisan lean, and other systematic components of the model; it is identifiable over time as many candidates appear repeatedly, whether they win or lose. The national trend η_t is the time-specific, realized exogenous shock measuring the national political environment for a given year: this models the familiar observation that some years are good for Democrats and others for Republicans, even if we do not know which is which ex ante. Alongside them, the model includes a midterm penalty term, M , which represents the strong tendency of the president's party to lose ground in midterm elections. It is folded into X_{it} in Equation 3. The model also includes

two party-premium terms, folded into X_{it} in the same way: A_d , a redistricting-decade-level term on incumbent-held seats measuring the extra value of holding a seat for the stronger party of the era, and E_d , the analogous term on voluntarily vacated seats. Because they vary across districts within an election year, they are identified separately from the year-level national trend η_t .

We also account for the fundamental variation in β , γ_c and η_t by estimating the corresponding standard deviations (σ_{β_k} , σ_γ , and $\sigma_{\eta,d(t)}$) for the distributions of these parameters. When constructing quantities of interest, we can then exclude, include estimated values, or integrate over the uncertainty in these distributions depending on the desired theoretical target.

For intuition, think about the vote on the logistic scale: let $y_{it} \equiv \ln[v_{it}/(1 - v_{it})] = \mu_{it} + \omega_{it} = X_{it}\beta_t + \gamma_{D(i,t)} - \gamma_{R(i,t)} + \eta_t + \omega_{it}$, with error term $\omega_{it} \equiv \ln[v_{it}/(1 - v_{it})] - \mu_{it}$, which Equation 2 indicates is t distributed. These expressions enable us to see, first, that national trends induce a positive covariance between any two districts i and j ($i \neq j$) for each election year t : $\text{Cov}(y_{it}, y_{jt} | X_{it}, X_{jt}, \beta_t) = \sigma_{\eta,d(t)}^2 > 0$. It also makes clear that the random candidate uniqueness terms $\gamma_{D(i,t)}$ and $\gamma_{R(i,t)}$ induce a positive covariance for election outcomes in any one district i at two times t and t' when the same candidate runs at both, over and above differences due to X : $\text{Cov}(y_{it}, y_{it'} | X_{it}, X_{it'}, \beta_t, \beta_{t'}) = \sigma_\gamma^2 (\mathbf{1}[D(i,t) = D(i,t')] + \mathbf{1}[R(i,t) = R(i,t')]) > 0$.⁵

2.2.3 Parameterizing the Uncertainty Structure and Concentration

The standard model assumes homoskedastic error variance across all races. The ALT already decomposes that constant variance, for estimating tail probabilities and concentration near the central tendency of the distribution, into scale and degrees-of-freedom parameters. We find that further flexibility is needed and so parameterize these distributional parameters in the LogisTiCC model, letting them depend on separate linear combinations of other covariates. We make this choice because some districts exhibit far more systematic, unmodeled variance than others. As it turns out, safe seats' future vote shares are

⁵Conditioning on β here simplifies the covariances without affecting the resulting inequalities by enabling us to drop $X_{it} \text{Var}(\beta_t) X_{jt}^\top$ and $X_{it} \text{Cov}(\beta_t, \beta_{t'}) X_{it'}^\top$, respectively.

relatively predictable from one election to the next, whereas a competitive seat exhibits a far greater tendency to surprise and confound predictions. Assuming a homoskedastic error variance structure would therefore overstate our certainty about the volatile pivotal races which determine control of Congress, while making under-confident predictions for uncompetitive elections.

The scale (4) is log-linear, with an intercept ζ and redistricting-decade fixed effects ζ_d . The same marginal distance from 50/50 (large for lopsided districts and near zero for even ones) is included in the specification for the degrees of freedom (5), which are built the same way, with a minimum value of 1 to give the distribution a finite mean. This means that, unlike the normal model, open seats are allowed to be both noisier and heavier-tailed than races with an incumbent on the ballot. The polarization terms let the data estimate the effect of competitiveness on the spread and the tail weight (e.g., whether contested ones have heavier tails). Finally, exponentiation keeps ϕ and the inner term of ν positive, because they are scale parameters.

The single-district state indicator in (4)–(5) is an indicator for states with only one district, such as Delaware or Alaska, and similarly one for all Senate elections. This is meant to capture the observation that voters in single-district states and Senate elections may have more informational content about the election (since they are statewide contests, highly covered in the media, covering many media markets) as opposed to one of the 52 elections in California.

2.2.4 The Model, Allowing for Uncontested Elections

In the standard approach, the vote in uncontested elections is typically recoded to fixed values such as $v_{it} = 0.75$ for Democrats running uncontested and $v_{it} = 0.25$ for Republicans running uncontested, or sometimes uncontested elections are deleted entirely. Uncontested races are not rare in the House: 21 percent of races in 1950 and 6.3 percent in 2020 (Supplementary Appendix S6). We instead formally distinguish between the observed vote v_{it} and the *effective vote* v_{it}^* , defined as the vote proportion that would have been observed if the election had been contested (e.g., King and Gelman, 1991); its logit is $y_{it}^* = \text{logit}(v_{it}^*)$. The effective vote is observed $v_{it}^* = v_{it}$ in contested elections but

unobserved if one party runs unopposed. We then view unobserved values of the effective vote (i.e., for uncontested elections) as unknown parameters to be estimated with all others. This approach includes all available information and appropriately accounts for all uncertainty in the imputation without imposing untenable assumptions hiding as coding decisions.

To model v_{it}^* when unobserved, we replace the outcome variable v_{it} in Equation 2 with the effective vote, and add a “censoring assumption”: *candidates running unopposed would have won even if the election were contested*. This assumption is intuitive, probably accounts for why the district was uncontested in the first place, and is a special case of the assumption made by Katz and King (1999). We then replace Equation 2 with

$$v_{it}^* \sim \text{ALT}(\mu_{it}, \phi_{it}, \nu_{it}), \quad (6)$$

and write the likelihood function for an election district that is fully contested as $\text{ALT}(v_{it} \mid \mu_{it}, \phi_{it}, \nu_{it})$, for a district where a Democrat runs uncontested as $\psi_{it} \equiv \int_{0.5}^1 \text{ALT}(v^* \mid \mu_{it}, \phi_{it}, \nu_{it}) dv^*$, and for a district where a Republican runs uncontested as $1 - \psi_{it}$. The integral implements the censoring assumption: the unopposed candidate would have won. In practice we bound this at 0.99 and 0.01 to avoid infinities at the bounds after the logit transformation.

The decision to include assumptions about uncontested districts in the model in a proper inference implies an additional complication: When the lagged effective vote is used as a covariate, it too can be unobserved, which necessarily adds a level of modeling complexity. A lag is also missing when the district was redrawn. We describe this feature, along with the full likelihood, in Supplementary Appendix S1.

2.2.5 Jointly Modelling House and Senate Elections

Scholars have applied many creative data analytic approaches to study the Senate and its sparse data, such as by averaging over years or Senate election cycles (e.g., Abramowitz, 1988; Highton, 2000; Carson, 2005). Our alternative approach is to first fit the same generative model to each house separately. Then, because the parameters from the two houses turn out to be statistically indistinguishable, we pool the models into one.

2.2.6 Priors

We estimate the model with priors to allow for Bayesian computation, but chosen to be weakly informative to insure the estimates are driven by the data (see Supplementary Appendix S1, Table S3). Supplementary Appendix S7 confirms that they are: for six structural parameters spanning the model, the posterior is far more concentrated than the prior, and doubling the scales of the dispersion and hierarchy priors leaves the held-out forecast errors unchanged while widening the intervals somewhat. When expanded fully, our model contains one candidate effect for each of 15,973 major-party candidates (3,598 of whom serve in Congress; the rest are challengers identified by name within their state), one latent lagged vote for each of the 5,555 races whose previous election was uncontested or interrupted by redistricting, and the 313 parameters that govern them—the slopes and their year-by-year movements, the national trends, the party-premium terms, and the scale and degrees-of-freedom coefficients—so the total number of parameters is $15,973 + 5,555 + 313 = 21,841$. But of course the prior shrinks the parameter space down to a much smaller effective number. Supplementary Appendix S8 reports the posterior estimates of the structural parameters and the distribution of the candidate effects.

3 Evaluation

We now evaluate both the standard linear-normal approach and our proposed LogisTiCC alternative. We do this by summarizing the models’ statistical properties (Section 3.1), comparing the probabilities of rare events from each approach to actual elections (Section 3.2), and studying the models’ credible interval coverage (Section 3.3). (We perform computation using Hamiltonian Monte Carlo using the no-U-turn sampler. See Supplementary Appendices S1 and S9.)

3.1 Statistical Properties

As political scientists have repeatedly shown, the linear-normal model can reveal important information about elections when its specification is close to correct. This standard approach is not a special case of the LogisTiCC, although it can be a good approxima-

tion in certain situations, and of course the LogisTiCC is designed to pick up features the normal cannot. As with all potentially “misspecified” models, point estimates from the linear-normal model will choose the distribution closest to the true data generation process (in the sense of the Kullback-Leibler information criterion; see White 1996) even if the data come from the LogisTiCC. In addition, if the linear specification is correct, both the normal and the LogisTiCC models will produce similar (and approximately consistent) estimates of (the same) β .

If the LogisTiCC is correct, the normal will usually give approximately unbiased, although highly inefficient, estimates of the β coefficients. Uncertainty estimates will also be incorrect. Other crucial quantities of interest, such as the probability of a candidate winning an election, will be statistically inconsistent under the normal but can be consistent with the LogisTiCC. Out of sample, the LogisTiCC is more accurate than the normal on every standard metric we compute, for point predictions and for full predictive distributions alike, in both chambers (Supplementary Appendix S10).

As we show below, a key problem with the linear-normal model is its assumptions of independence across districts, candidates, and time. In contrast, the LogisTiCC allows for dependence among elections held in the same district at different times and among elections held in different districts on the same day. Correcting for this false precision leads to appropriately larger credible intervals but also has a chance of getting distributional quantities correct, such as the probability of winning elections in competitive races. Although the differences vary over applications, we find that the ratio of the nominal width of LogisTiCC-to-normal credible intervals is about 1.07 for district-level predictions (Supplementary Appendix S11) and about 5.2 for aggregate predictions of interest such as the vote for the median house seat.

3.2 Rare Event Probabilities

Although the normal model often performs well for estimating quantities estimated in the literature such as regression coefficients, we now pressure test these models by looking at the most difficult cases, in particular the probability of extraordinarily rare events. Although the literature does not calculate these probabilities (as most do understand that the

normal model would not be trusted for these purposes), they are a particularly useful standard when trying to establish a single data generation process applicable to all US election data. For illustration, we use the notion of *moral certitude* from the Enlightenment, which is that events with probabilities smaller than 1 in 10,000 should be disregarded. (Because demographers of the time observed that the probability of a healthy person dying in the next day was smaller than 1 in 10,000 and does not seem to affect people’s behavior in their daily lives, people act as if they are “morally certain” that these rare events will never occur; see Kavanagh 1990; Buffon 1777.) Updating this (quaint) idea, we make predictions for all elections in our dataset and count the number of elections for which the vote proportion observed out-of-sample appears outside a 99.99% (i.e., $1 - 1/10,000$) forecast credible interval. If the interval is correct, we should observe about 1 in 10,000 outside the interval.⁶

Figure 1 gives a count of these extraordinarily rare events (on the vertical axis) by election year (on the horizontal axis) and for the normal model (in gold) and the LogisTiCC (in black). As we forecast 14,478 contested elections one election ahead, we would expect to see 1.45 violations of “moral certitude”. The normal model (gold line) produces 62 violations; the LogisTiCC model generates 2, which is about what we would expect to see if it is the data generating process. Tightening the threshold to one in 100,000 changes nothing: the normal model still produces 30 violations and the LogisTiCC none.

To emphasize how far off these predictions are, consider some of the points in the figure annotated with the exact probability of seeing the data observed under the specified model. For the worst predictions, we observe so many rare events that the model would lead us to expect to see this many surprises once in every 4.8 quintillion election years (or $\sim 350,000,000$ times the estimated age of the universe). The late political scientist Richard McKelvey was fond of arguing that a fix for over-claiming in empirical work would be to require anyone reporting a p-value to take a bet with the implied odds (i.e., the reciprocal

⁶Our model architecture is autoregressive by construction and so standard out-of-sample validation methods such as leave-one-out cross-validation are not appropriate. We therefore adopt a leave-future-out validation benchmark for out-of-sample testing: for $t = 1950, 1952, \dots, 2024$, we train our model on all elections in our data panel up to year $t - 2$, and then forecast election year t from there. We begin these leave-future-out tests at $t = 1950$ for the House and $t = 1960$ for the Senate to provide enough training data. We show the similarity of these evaluation methods and almost identical results in Figure S11.

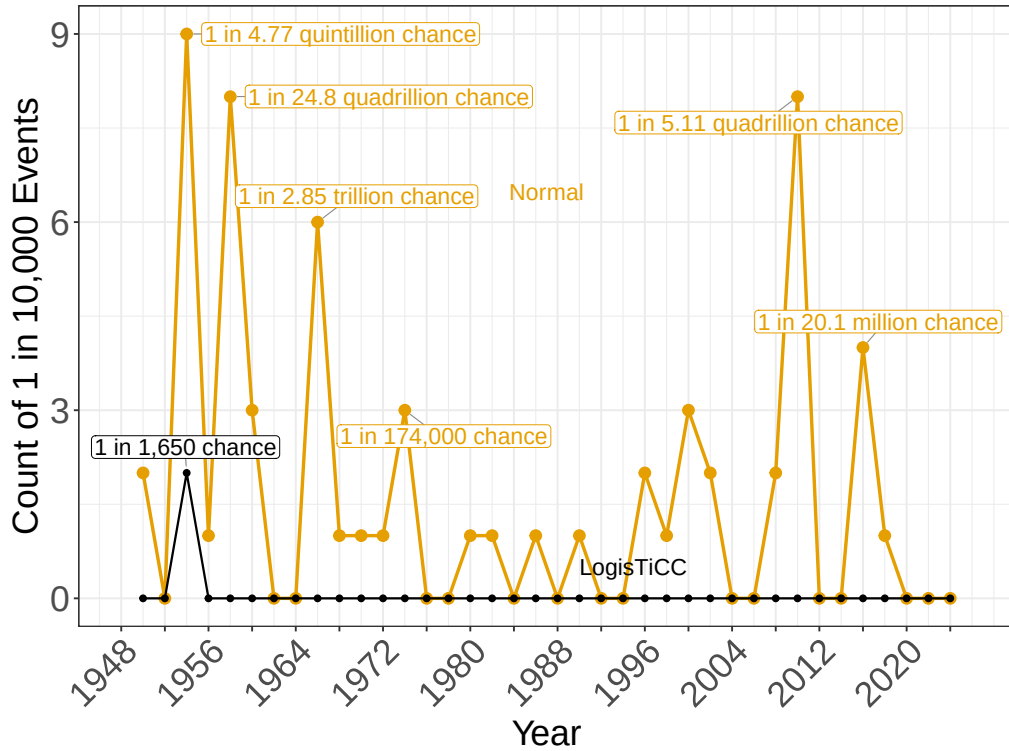


Figure 1: Moral Certitude: Count of elections whose observed vote falls outside the 99.99% one-election-ahead credible interval, for each election year (with selected points labeled with the probability each model gives of seeing this many 1-in-10,000 events). Separate calculations appear for the normal model (in gold) and our proposed LogisTiCC model (in black).

of the p-value to one) against someone finding evidence to the contrary. Using this logic, a one dollar bet against the linear-normal model’s claimed level of certainty would give an equal chance of winning quintillions of times more money than exists in circulation in all the world’s currencies.

Thus, for this measure of extraordinarily unlikely events, the out-of-sample performance of our proposed model vastly exceeds that of the standard approach. We now show that this result is general in that the probabilities from our model, but not the normal, are well *calibrated*, meaning that for example when the model predicts that a certain event will occur with a 0.3 probability, that event actually occurs in about 3 of every 10 elections, and so on. Thus, for each election and model, we first compute the (out-of-sample) probability of a competitive outcome (which we define here as $v_{it} \in [0.45, 0.55]$). We then sort these probabilities into bins, $[0, 0.1]$, $(0.1, 0.2]$, $(0.2, 0.3]$, ..., separately for each

model, and plot them in Figure 2, as follows. For each model, the horizontal coordinate of a dot is the average of the estimated probabilities of elections in a bin and the vertical coordinate as the fraction of out-of-sample elections in the same bin that are in fact observed to be competitive. Dots for a perfectly calibrated model should fall approximately on the 45 degree line.

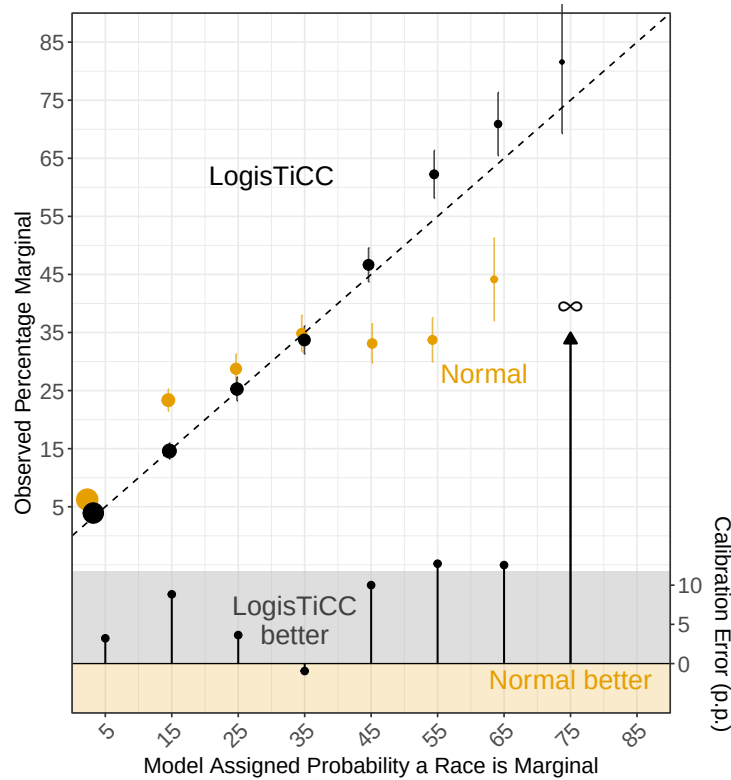


Figure 2: Calibration for Marginal Districts: Predicted out-of-sample probabilities (horizontally) by observed frequencies (vertically). Calibration error is reported for each corresponding bin in the lower panel.

As Figure 2 demonstrates, the dots computed from the LogisTiCC bins (in black) are fairly close to the 45 degree line, and hence well calibrated. In contrast, those from the normal (in gold) substantially deviate from the 45 degree line of equality as the predicted probability of a competitive election grows. We summarize the results from the model in the bottom section, which gives the differences, above the line indicating that the LogisTiCC is better. In sum, the normal model fails most dramatically in elections that are most politically important, the competitive ones. In fact, for bin with the highest probability of a competitive election, the normal, which discriminates less well, gives no predictions at

all.

We now go further and bin election contests by the predicted effective vote and plot each bin's mean prediction against its mean realized vote out of sample (see Figure 3). Both models perform well on this metric, but the standard model fares worse in the important competitive election range (differences appear in the inset). Supplementary Appendix S12 (Figure S12) shows that the LogisTiCC also calibrates well within each redistricting decade.

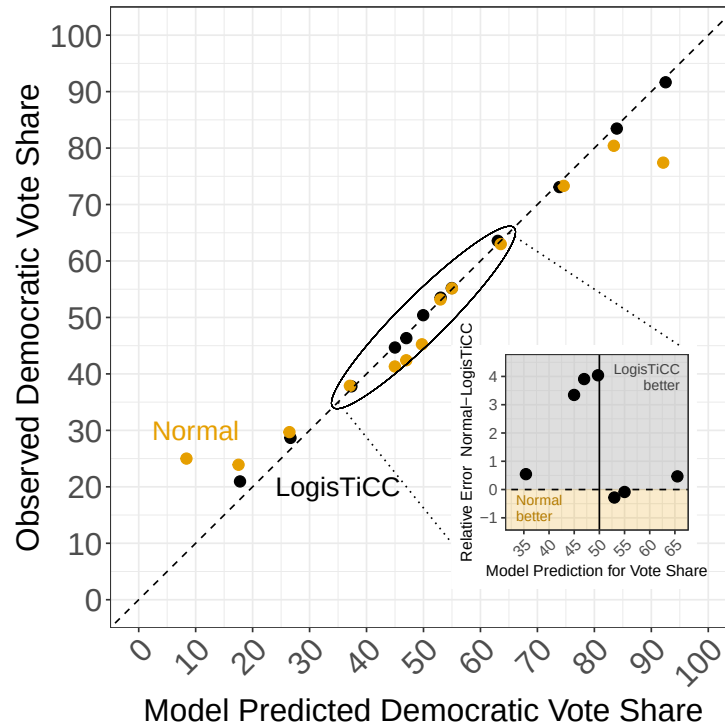


Figure 3: Calibration of Vote Share. Out-of-sample contests binned by predicted effective vote, horizontally, against mean realized vote, vertically, with an inset giving the standard model's excess absolute error over the LogisTiCC near the threshold. Both models track the 45° line; the standard model's shortfall is concentrated near one-half.

3.3 Coverage

We now study, in three ways, the properties of credible intervals computed from the two models. First, the red stars in Figure 4 gives a time series of one of the most consequential quantities of interest in US politics — control of the majority in the House of Representatives, as determined by the Democratic proportion of the vote of the median seat. For each

year and model, we fit both models to all elections (using data up to two years earlier) and compute a point forecast and 95% out-of-sample credible interval for it. These appear in gold for the normal and black for the LogisTiCC. To interpret these results note that a t -based interval has both fatter tails to accommodate surprises and more concentration of density near the mean than the normal, making a mean prediction more informative than a normal even if the normal were correct. In addition, as we showed above, the LogisTiCC intervals are accurate whereas the normal intervals are overconfident. The results are summarized in the right panel. Under the models, we would expect 1.90 misses out of 38 for LogisTiCC and indeed its predictive credible interval captures the observed outcome in all but two of the 38 years. In contrast, under the model, we would expect 1.60 misses for the normal (we have more elections for the LogisTiCC model because we impute redistricting years), but it misses 28 of 32.

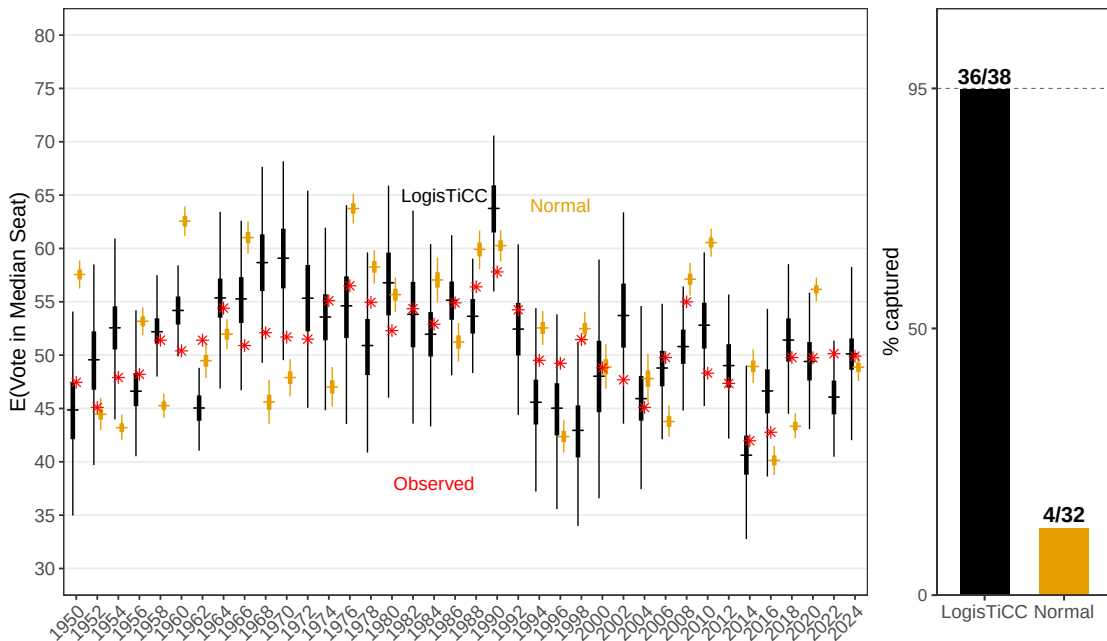


Figure 4: Expected Vote Share of the Median House Seat (95 Percent Credible Interval)

Second, for each model, we compute a 95% credible interval around every individual district's (out of sample) vote share and tally up the percentage of districts that interval captures. Our results appear in Figure 5, with election year on the horizontal axis and percent coverage on the vertical axis (again with normal in gold and LogisTiCC in black).

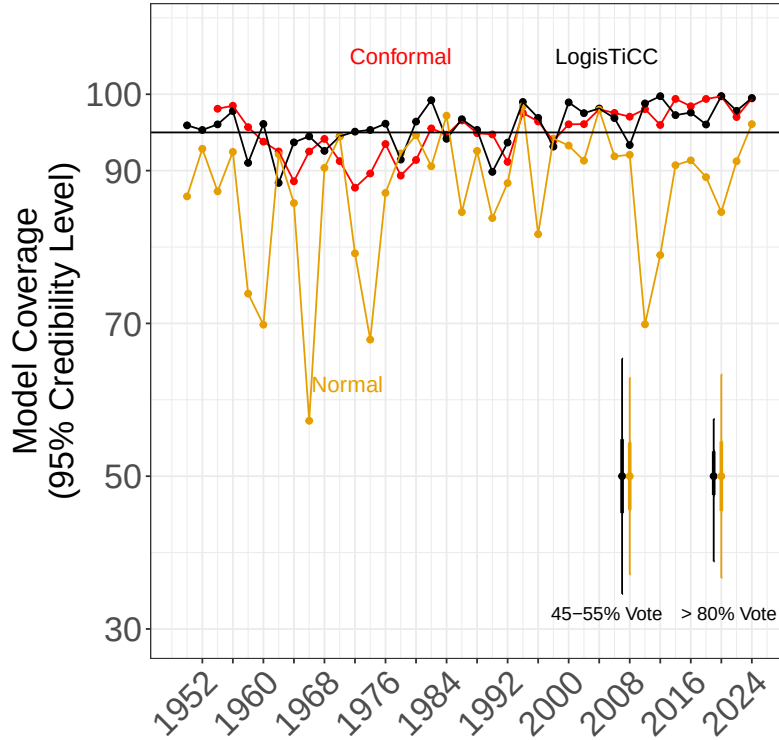


Figure 5: Coverage under Each Model at the 95 Percent Level

A properly calibrated model should capture 95% of districts which, aside from estimation error, should be at the flat black line near the top of the figure. This is close to the case for the LogisTiCC. In contrast, the normal interval substantially deviates from capturing 95% of the elections in all but a few years. The bottom right of the same panel gives average confidence interval sizes for two regions of the vote, showing that in one region the normal intervals are wider and in the other LogisTiCC is wider — although in both the LogisTiCC is accurate but normal is not.

Finally, we evaluate our specific distributional assumption (a compound error term with random effects and an additive logistic t distribution). To do this, we use methods of “conformal inference” that offer guarantees of accurate distribution-free finite sample coverage even under model misspecification, for any predictive model (Vovk, Gamerman, and Shafer, 2005). (Intuitively, the method works by computing confidence intervals based on errors from previous years’ forecasts, assuming that the data generation process is exchangeable conditional on the covariates.) We thus add to Figure 5 confor-

mal confidence intervals (in red). We first note that the conformal intervals have accurate coverage, as designed, which we can see as the red line varies around the flat 95% line across the years. More relevant for our purposes is the comparison between the fit of the red and black lines to the 95% line. This comparison indicates that the LogisTiCC has approximately the same high quality coverage as these distribution-free intervals. These results thus provide evidence for the veracity of our distributional assumptions and for our Bayesian model as a generative model of US congressional elections data.⁷

4 Extension to the US Senate

We begin by re-estimating the model with every parameter interacted with an indicator for whether the observation came from the Senate or House. Figure 6 includes these interactions and reveals no systematic difference between the two legislative chambers, and none larger than a few percentage points of vote share. One of the nine differences excludes zero, but it is small in substantive terms—one point on the midterm penalty. Although the House and Senate have many structural legislative differences, electorally they follow analogous rules. Both feature first-past-the-post, single-member districts elected to fixed terms in a federal legislature. The similarity of the electoral institutions and the statistical evidence here are consistent with the claim that House and Senate elections share an underlying data-generation process. Given these results, we conclude that the chambers do not differ enough to support separate parameters and in our final runs we include them both together with common parameters.

Finally, we compute every evaluation for the Senate exactly as we did for the House in Section 3, from the same jointly estimated model, with no chamber-specific parameters (see Figure 7). The results match the House in kind: the one-election-ahead tail counts, the interval coverage, and the decade-by-decade calibration reproduce the House pattern, while on competitive-race and vote-share calibration the two models are close, with the error differences in panels (b) and (c) of Figure 7 within a few points on either side. The clearest evidence for that pooling is structural. We do not distinguish between observa-

⁷Supplementary Appendix S12 (Figure S11) shows that LogisTiCC coverage that we have evaluated in this section out of sample also works approximately as well in sample.

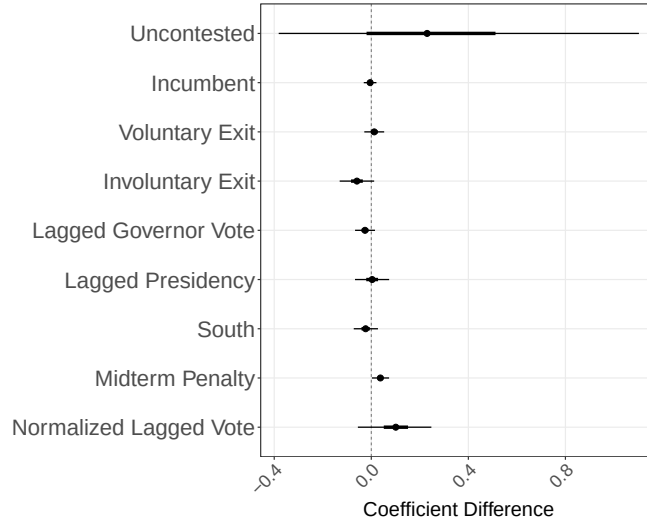


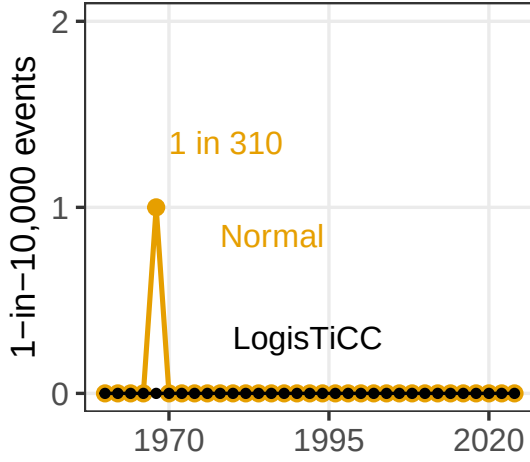
Figure 6: Little Difference Between House and Senate. Each structural parameter of the model is estimated with Senate-specific interaction terms to compare against a House baseline. Points are posterior medians and bars are credible intervals. Differences are Senate minus House on the logit scale of the two-party vote, per unit of the covariate; near an even race, 0.04 on this scale is one percentage point of vote share, so the one interval that excludes zero amounts to about +1 point (a smaller midterm penalty). Lagged vote is normalized to account for the differences in Senate (6 years) and House (2 years) terms.

tions from the House and the Senate in our training data: all predictions for both chambers are based on the same set of structural parameters.

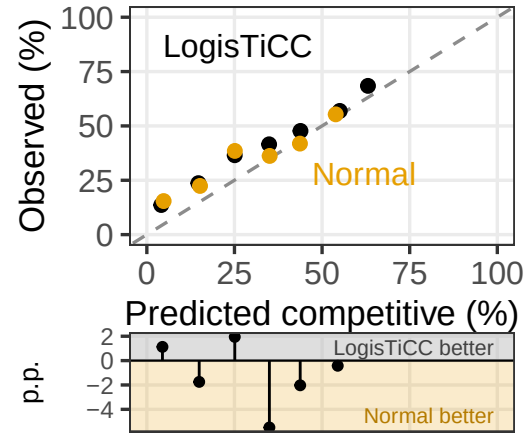
5 Generatively Accurate Descriptive Summaries

An advantage of a generative accurate statistical model is that researchers can use it to compute a vast array of different quantities of interest to social scientists, including descriptive, causal, counterfactual, and others. We outline here some of the quantities that can be estimated from generatively accurate models, such as ours, and explain how they can be used to enrich political science research (with Supplementary Appendix S13 placing this approach in the broader history of the generative modeling, causal inference, and machine learning literatures).

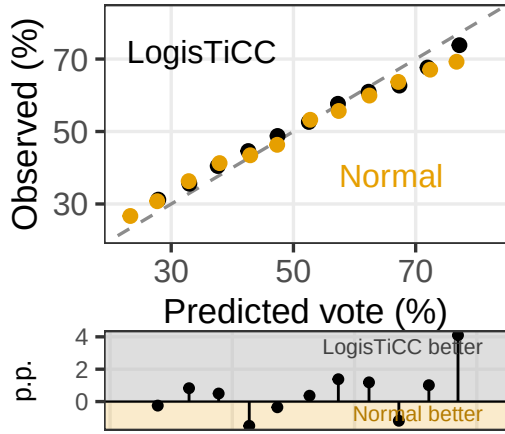
As inference is simply “using facts we know to learn about facts we do not know” (King, Keohane, and Verba, 1994), we characterize the types of quantities we may wish to estimate by first detailing both the “unknowns” that may be of interest and then the “knowns” we have available to condition on. We characterize the unknowns in three



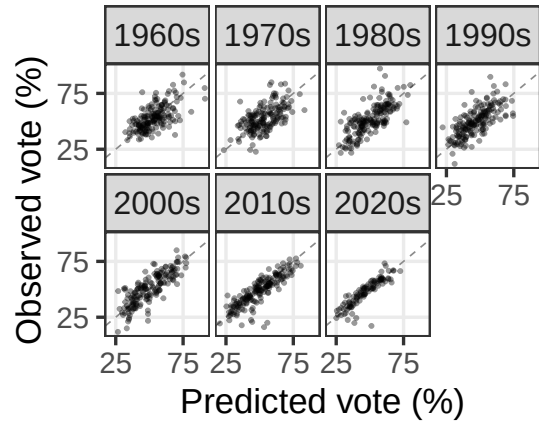
(a) Moral certitude



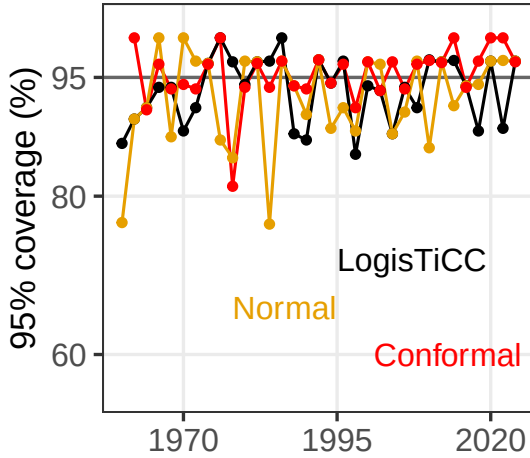
(b) Calibration: competitive races



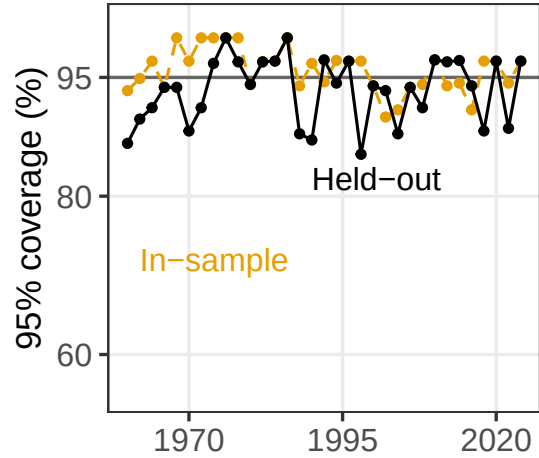
(c) Calibration on the vote scale



(d) Calibration by redistricting decade



(e) 95% interval coverage



(f) Coverage: held-out and in-sample

Figure 7: US Senate. Evaluations of Section 3 repeated for the Senate: (a) count of elections outside a 99.99% interval; (b) predicted against observed probability a race is competitive; (c) predicted vs observed vote share; (d) out-of-sample calibration within redistricting decades; (e) 95% interval coverage for the LogisTiCC, Normal, and one-step-ahead conformal; (f) LogisTiCC held-out coverage against the in-sample counterfactual.

ways. First, the *location* of an unknown is where the values are of the outcome variable that we want to know. This may involve a “forecast”, i.e., into the future; “farcast,” i.e., to an election in the present or past not in our dataset (such as for a different office or country); “nowcast”, i.e., to unobserved features of elections in our dataset (such as the posterior distribution for a district vote, only one value of which is observed, as in posterior predictive checks); or even a “faroutcast,” which refers to values of the outcome variable under counterfactual conditions (such as if no incumbents had run). Second is the *level* of aggregation of the quantity of interest, such as for district-, state-, regional-, or national-level statistics, or features of non-geographic groupings like all Democratic districts or all those without an incumbent. Finally, our quantities of interest involve a *concept*, such as partisan bias, electoral responsiveness, the probability that an incumbent will lose, the expected vote in a district, or the district vote of the median legislator.

Quantities of interest always condition on three types of features that are either known or, in the case of counterfactuals, assumed known. These include (1) the choice of covariates and their values for unknown quantities; (2) keeping, removing, or zeroing out random effects; and (3) keeping, removing, or adjusting surprises (such as to focus only on the expected value or other features of the posterior). We explain how to make decisions for choosing quantities of interest, and how to mix and match the location, level, and concept of the unknowns with the covariates, random effects, or error term surprises to condition on.

Consider estimating the probability that a Democrat wins a particular district i in election year t . The simplest case is a “nowcast”. Here we are interested in the ex ante probability that the Democrat would win this district election, which in fact we have already observed (ex post). To do this, we set the values of X to their observed district values. For example, suppose the lagged vote for the Democrat incumbent is 74%. We might then consider setting $\beta_{tk} = \hat{\beta}_{tk}$ for all k covariates, and $\gamma_{D(i,t)} = \hat{\gamma}_{D(i,t)}$, $\gamma_{R(i,t)} = \hat{\gamma}_{R(i,t)}$, and $\eta_t = \hat{\eta}_t$ for the random effects. Of course, we do not know any of these numbers for certain ex ante and, therefore, we would choose instead to include estimation uncertainty in our estimate of the probability that the Democrat would win this district. Thus, instead

of fixing these parameters at their point estimates, we draw them from their posteriors, centered on the estimated values. Then, suppose we take 1,000 draws from this posterior; to generate our model-based “nowcast” of v_{it} , we then multiply each of these draws by the relevant X_{itk} and add the draws of the candidate effects and national trend and draws of Student- t noise. We then run this sum through the inverse logit function to get a hypothetical draw on the scale of votes. This generates 1,000 hypothetical draws of v_{it} . To estimate the probability a Democrat wins the district, we then simply count up the fraction of the draws greater than 0.5.

For forecasting, the choices about how to construct generatively accurate descriptive statistics is more flexible, and thus more complicated. Consider the simplest case. If all we want is a one-election-ahead forecast, we still have a number of important decisions to make. For example, how do we set the covariate values? If it is the next election, we have observed the lagged vote, so that is straightforward to use, but what about incumbency? Do we know yet if the current incumbent will run again? If so, we could use that value. But if we are making the forecast well before the election, and do not know this yet, what value should we use? We could assume they all run, or some randomly selected proportion run again. Perhaps, however, it is better to consider what would happen if the district were open? Regardless, the analyst must choose some value relevant to the question at hand, and must realize that this choice changes the question we are answering. In fact, the differences in forecasts across these assumptions may be of considerable substantive value.

We also have important decisions about the candidate effects, $\gamma_{D(i,t)}$ and $\gamma_{R(i,t)}$. If this were only one election ahead, and we do not think much else has changed, then we may want to fix $\gamma_{D(i,t)} = \hat{\gamma}_{D(i,t)}$ and $\gamma_{R(i,t)} = \hat{\gamma}_{R(i,t)}$ as we did in our “nowcast” above. A new candidate’s effect is drawn from the implied candidate distribution, $\mathcal{N}(0, \sigma_\gamma^2)$. However, we surely do not know β_{tk} . So instead we need to use draws of it from $\beta_{tk} \sim \mathcal{N}(\hat{\beta}_{t-2,k}, \sigma_{\beta_k}^2)$, where $\hat{\beta}_{t-2,k}$ is the estimated value for the lag of β_{tk} and σ_{β_k} is the standard deviation for the distribution describing the random walk for β_k . This will add additional uncertainty to our forecast, but is otherwise similar to our “nowcast”. And

we are unlikely to know the national trend, so we must make a parallel choice about η_t . Given all these assumptions, we can then generate our hypothetical election draws and calculate the fraction of times the Democrat wins as our prediction as a probability.

Perhaps the most difficult set of choices comes from in making a “faroutcast”, as for example, when we want to forecast what would happen in a new legislative map following the implementation of a proposed (or perhaps recently passed) redistricting plan. First, we generally will not know where incumbents will be, but can often make some educated guesses. Alternatively, we might assume all seats are open to obtain a baseline probability that a Democratic candidate could win the seat. Harder yet, is what to do about lagged vote, which is giving the model our measure of the normal vote, the lagged effective vote. We could use precinct level returns for the previous election, subtract out the incumbency advantage and uncontestedness, aggregate into the new districts, and then add back in the incumbency advantage for districts where the decisions of incumbents and challengers is known. Or perhaps we could use presidential vote, or some average of statewide votes re-aggregated in the new district map. These constructed measures would be needed in the original model or in a separate model that imputes lagged vote from some statewide measures. Also, as with our forecast, we do not know β_{tk} , $\gamma_{D(i,t)}$, $\gamma_{R(i,t)}$, or η_t . And as before we could then generate our hypothetical election draws and calculate the fraction of times the Democrat won.

The flexibility of the model easily enables one to calculate even more sophisticated quantities of interest. For example, one of the largest sources of uncertainty in election predictions is the national trend. We can thus draw η_t directly from its posterior. Alternatively, we can model the parameters of the η_t prior with national-level covariates, such as unemployment, presidential approval, or whether the country is at war. Yet another option is to fix it at the value of some previous election that seems similar to the current one.

By combining the location, level, and concept for a quantity of interest and fine tuning by making choices about the covariates, random effects, and surprises, accurate generative models like the one we describe here can reveal a vast amount about American elections, far richer than any one specific estimate or data analysis on its own.

6 Concluding Remarks

Commonly used models of district-level election results have enabled political scientists to learn a wide variety of information about American legislative democracy. But the observable implications of these models fail spectacularly and regularly in ways that should almost never happen. We build on existing approaches by adding substantive features of elections political scientists have learned over the years, and building on new statistical and computational technology not previously available. We validate our approach with extensive out-of-sample (and distribution-free) tests in 15,582 district-level elections. Our generative model is general in that it can be used, with the appropriate additional assumptions and covariates when necessary, to estimate almost any quantity of interest in the literature, and others, all with calibrated (i.e., accurate) probabilities and honest uncertainty intervals.

Software we make available includes the model, the data, and the held-out forecasts together, so that researchers can construct and compute quantities of interest of their choice. From this data and software, researchers can calculate a vast array of estimates of scholarly interest, all validated from this single generative model. We look forward to seeing what we all learn.

References

- Abramowitz, Alan I. (1988). “Explaining Senate Election Outcomes”. In: *American Political Science Review* 82.2, pp. 385–403. DOI: [10.2307/1957392](https://doi.org/10.2307/1957392).
- Beck, Nathaniel and Jonathan N. Katz (2007). “Random Coefficient Models for Time-Series–Cross-Section Data: Monte Carlo Experiments”. In: *Political Analysis* 15.2, pp. 182–195. DOI: [10.1093/pan/mpl001](https://doi.org/10.1093/pan/mpl001).
- Buffon, George Louis Leclerc de (1777). “Essai d’arithmétique morale”. In: *Euvres philosophiques*.
- Carson, Jamie L. (2005). “Strategy, Selection, and Candidate Competition in U.S. House and Senate Elections”. In: *The Journal of Politics* 67.1, pp. 1–28. DOI: [10.1111/j.1468-2508.2005.00305.x](https://doi.org/10.1111/j.1468-2508.2005.00305.x).
- Clinton, Joshua, Simon Jackman, and Douglas Rivers (2004). “The Statistical Analysis of Roll Call Data”. In: *American Political Science Review* 98.2, pp. 355–370. DOI: [10.1017/S0003055404001194](https://doi.org/10.1017/S0003055404001194).

- Gelman, Andrew and Gary King (May 1994). “A Unified Method of Evaluating Electoral Systems and Redistricting Plans”. In: *American Journal of Political Science* 38.2, pp. 514–554. URL: j.mp/unifiedEc.
- Gill, Jeff (2008). *Bayesian Methods: A Social and Behavioral Sciences Approach*, 2nd edition. Chapman & Hall/CRC.
- Giroso, Federico and Gary King (2008). *Demographic Forecasting*. Princeton: Princeton University Press. URL: j.mp/dsmooth.
- Goplerud, Max (2022). “Fast and Accurate Estimation of Non-Nested Binomial Hierarchical Models Using Variational Inference”. In: *Bayesian Analysis* 17.2, pp. 623–650. DOI: [10.1214/21-BA1266](https://doi.org/10.1214/21-BA1266).
- Highton, Benjamin (2000). “Senate Elections in the United States, 1920–94”. In: *British Journal of Political Science* 30.3, pp. 483–506. URL: <http://www.jstor.org/stable/194005>.
- Jackman, Simon (2009). *Bayesian Analysis for the Social Sciences*. Chichester: Wiley.
- Katz, Jonathan N, Gary King, and Elizabeth Rosenblatt (2020). “Theoretical foundations and empirical evaluations of partisan fairness in district-based democracies”. In: *American Political Science Review* 114.1, pp. 164–178. URL: GaryKing.org/symmetry.
- Katz, Jonathan N. and Gary King (Mar. 1999). “A Statistical Model for Multiparty Electoral Data”. In: *American Political Science Review* 93.1, pp. 15–32. URL: bit.ly/mtpty.
- Kavanagh, Thomas M (1990). “Chance and Probability in the Enlightenment”. In: *French Forum*. Vol. 15. 1, pp. 5–24.
- King, Gary and Andrew Gelman (Feb. 1991). “Systemic Consequences of Incumbency Advantage in the U.S. House”. In: *American Journal of Political Science* 35.1, pp. 110–138. URL: bit.ly/SystCs.
- King, Gary, Robert O. Keohane, and Sidney Verba (1994). *Designing Social Inquiry: Scientific Inference in Qualitative Research*. Princeton: Princeton University Press. URL: bit.ly/gKroKsV.
- Lax, Jeffrey R. and Justin H. Phillips (2009). “How Should We Estimate Public Opinion in the States?” In: *American Journal of Political Science* 53.1, pp. 107–121. DOI: [10.1111/j.1540-5907.2008.00360.x](https://doi.org/10.1111/j.1540-5907.2008.00360.x).
- Linzer, Drew A. (2013). “Dynamic Bayesian Forecasting of Presidential Elections in the States”. In: *Journal of the American Statistical Association* 108.501, pp. 124–134. DOI: [10.1080/01621459.2012.737735](https://doi.org/10.1080/01621459.2012.737735).
- Martin, Andrew D. and Kevin M. Quinn (2002). “Dynamic Ideal Point Estimation via Markov Chain Monte Carlo for the U.S. Supreme Court, 1953–1999”. In: *Political Analysis* 10.2, pp. 134–153. DOI: [10.1093/pan/10.2.134](https://doi.org/10.1093/pan/10.2.134).
- Park, David K., Andrew Gelman, and Joseph Bafumi (2004). “Bayesian Multilevel Estimation with Poststratification: State-Level Estimates from National Polls”. In: *Political Analysis* 12.4, pp. 375–385. DOI: [10.1093/pan/12.4.375](https://doi.org/10.1093/pan/12.4.375).
- Riker, William H (1990). “Heresthetic and rhetoric in the spatial model”. In: *Advances in the spatial theory of voting* 46, p. 50.
- Shepsle, Kenneth A (2003). “Losers in politics (and how they sometimes become winners): William Riker’s heresthetic”. In: *Perspectives on politics* 1.2, pp. 307–315.

- Shor, Boris, Joseph Bafumi, Luke Keele, and David Park (2007). “A Bayesian Multilevel Modeling Approach to Time-Series Cross-Sectional Data”. In: *Political Analysis* 15.2, pp. 165–181. DOI: [10.1093/pan/mpm006](https://doi.org/10.1093/pan/mpm006).
- Steenbergen, Marco R. and Bradford S. Jones (2002). “Modeling Multilevel Data Structures”. In: *American Journal of Political Science* 46.1, pp. 218–237. DOI: [10.2307/3088424](https://doi.org/10.2307/3088424).
- Vovk, Vladimir, Alexander Gammernan, and Glenn Shafer (2005). *Algorithmic learning in a random world*. Springer Science & Business Media.
- Western, Bruce (Oct. 1998). “Causal Heterogeneity in Comparative Research: a Bayesian Hierarchical Modelling Approach”. In: *American Journal of Political Science* 42.4, pp. 1233–1259.
- White, Halbert (1996). *Estimation, Inference, and Specification Analysis*. New York: Cambridge University Press.